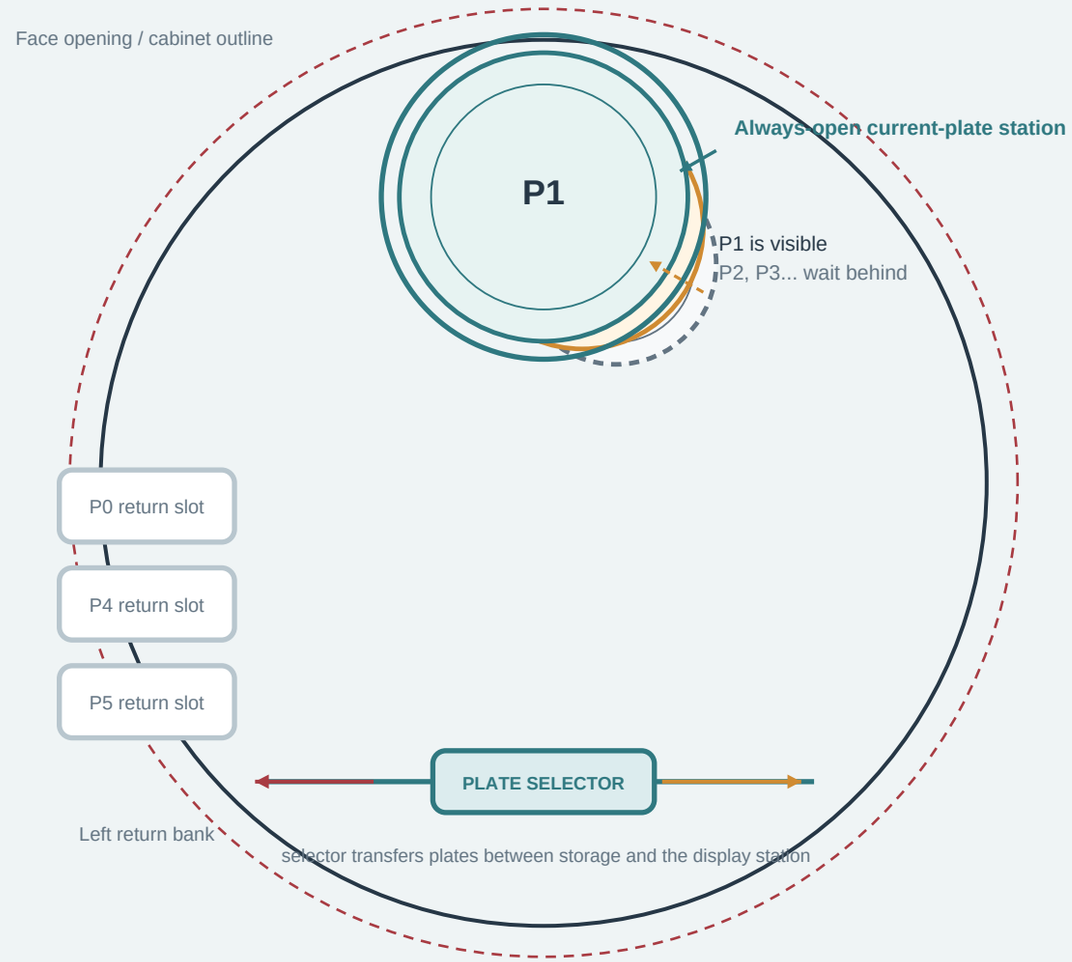
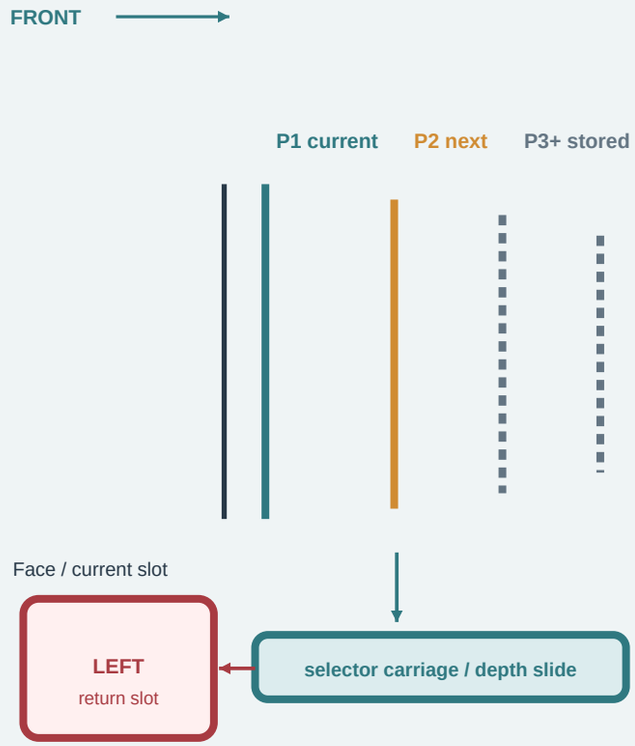
Portrait plate mechanism

Working concept based on the described exchange sequence; schematic only, no depth scale.

FRONT ELEVATION



SIDE SECTION | PLATES IN DEPTH



P1 slides back onto the selector, then moves left into its own slot.

The next plate is picked from the rear stack and carried forward.

Design intent

The display station is reserved for the current portrait and stays open.

The surrounding mechanism closes only after the new plate is aligned there.

Color key: teal = current plate / carrier; gold = selected next plate; red = return movement; dashed = stored plate or guide path.

Proposed geometry is illustrative. Plate count, selector track shape, and exact pivot locations remain to be set.

Change sequence

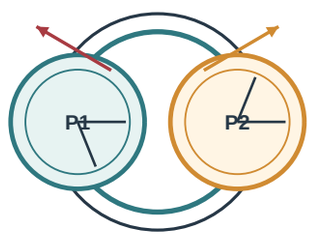
One plate exchange from the displayed portrait to the newly selected portrait.

1 Display P1

next plate rests behind

P1 occupies the reserved front slot; other portraits wait behind it.

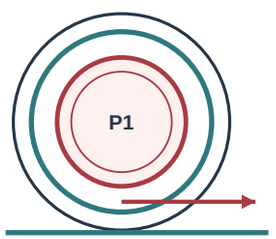
2 Open for clearance



rotate away from each other: 68 deg

The two moving plate members rotate away from each other by 68 deg.

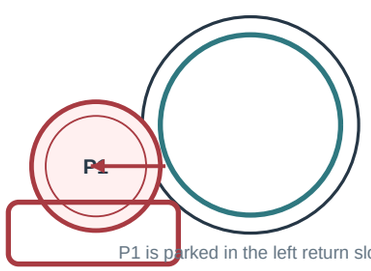
3 Retract current plate



P1 retracts onto selector carriage

P1 slides backward onto the plate-selector carriage.

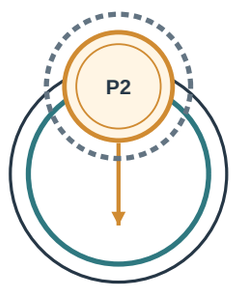
4 Return P1 left



P1 is parked in the left return slot

The selector carries P1 into its assigned storage slot on the left.

5 Present next plate



selector brings P2 forward

The selector takes P2 from the queue and brings it to the front.

6 Align and close

P2 aligns; surrounding assembly closes

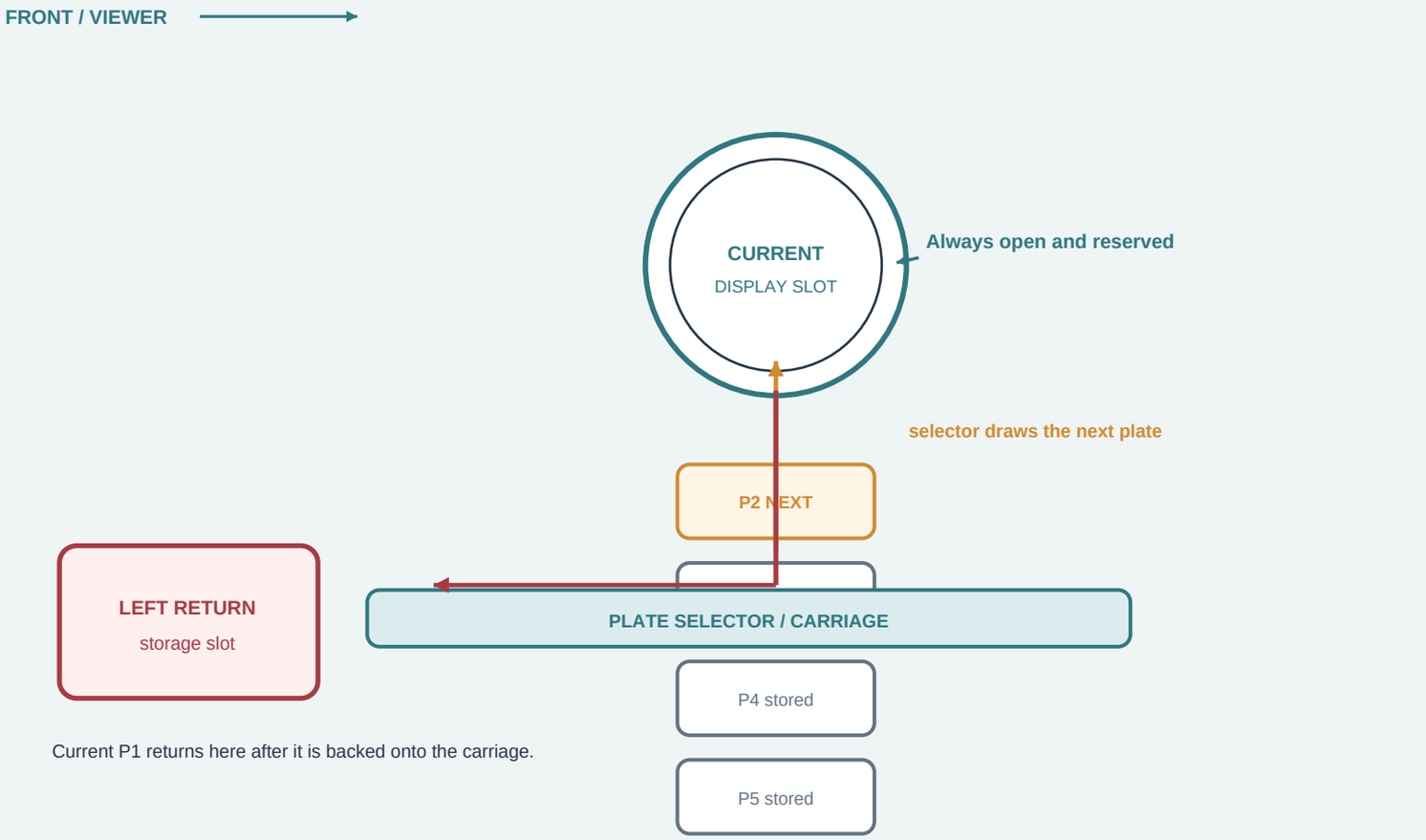
Once P2 aligns with the current slot, the surrounding assembly closes.

The current-plate opening remains reserved throughout; it is not used as a storage position.

Selector routing and swing clearance

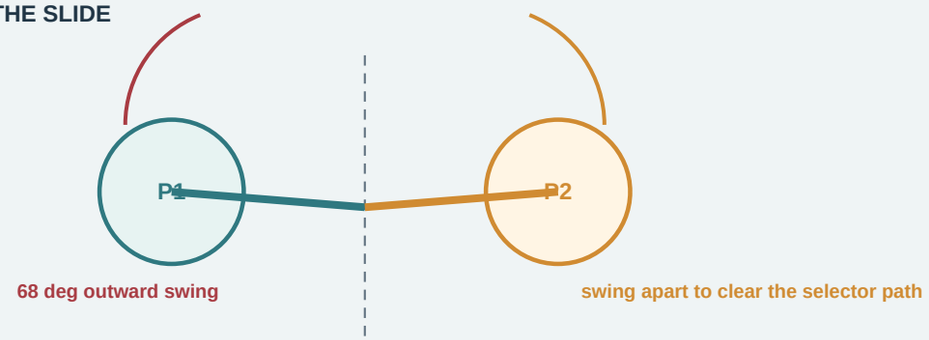
Plan-view routing sketch. Exact pivots and guide geometry are intentionally left open for review.

TOP-DOWN / DEPTH ROUTING



Current P1 returns here after it is backed onto the carriage.

OUTWARD SWING BEFORE THE SLIDE



MECHANISM RULES

1. One permanent display station

The current portrait always returns to the same front slot. Keep this station clear of storage positions.

2. Separate swing from indexing

Swing the two members apart first. Only then move a plate through the selector route.

3. Return before advancing

The outgoing portrait goes to its left slot before the selector presents the next one.

4. Close only on alignment

The new plate must sit squarely in the display station before the surrounding assembly closes.

STILL TO SET

- Pivot axis and exact swing members
- Plate count and left-slot capacity
- Carrier / selector path and drive
- Depth, tolerances, and collision checks

Schematic interpretation: 68 deg is shown as the outward separation motion; exact pivot geometry awaits the next design pass.